Supplementary Material
Trajeform: Embedding Crowd Data

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This supplementary material mainly contains additional figures for more detailed breakdown of the methodology, and more instances of results. A supplementary video can be accessed via: *https://youtu.be/g3ugxktiWNs*

I. MPACT Background

For experiments conducted in the paper, we mainly use three simulator-generated datasets which we call RVO, CCP, and MPACT. The former, RVO [1] is an early simulator, highly effective in reactive behaviours such as collision avoidance and steering. Configurable Crowd Profiles (CCP) was introduced by Panayiotou et al. [2] and it is a reinforcement learning based simulator, that achieves a mixture of behaviours through weighted behaviour parameters (profiles), i.e., goal-seeking, grouping, interaction with Points of Interest (POIs), and collision avoidance.

MPACT [3] introduces a behaviour optimisation model that finds the set of behaviour parameters best describing the inputs; inputs are given as reference trajectories. These sets behaviour parameters comprise a profile, as this is defined by the collection of four behaviour weights given by the modCCP simulator (a modified version of the CCP simulator, introduced in the same paper [3]). In particular, these four weights correspond to goal-seeking, grouping behaviours, interaction with POIs, and the connectivity weight which is meant to describe how "connected" agents that move together appear. Hence the inputs to modCCP are such profiles, and the outputs are the simulated trajectories subject to those behaviours, while for MPACT (optimisation framework), the inputs are sets of reference data and the outputs are the optimised profiles (which are then passed through modCCP to also get the respective simulated trajectories). For more information, the reader is referred to the original research papers.

II. Methodology Details

Figure 1 provides a breakdown of the amount of agent trajectories found in each of the real, established trajectory datasets (Flock, Students, Zara, ETH). Figure 7 shows the components considered when building the preprocessing module visually, using example trajectories. Figure 2 illustrates examples of sample points from the learned latent space, from three sampling methods, for sampling size of 1k.

III. Complementary Analysis

A. Model Variants: "Hold-out" Datasets

Figure 9 provides a visualisation of the "hold-out" models experiment. It shows, for each of the test data (Students, Zara, Flock, ETH) their latent projections in the full model (trained on the full four datasets), against their latent projections on the latent spaces of the model variants that were trained on the datasets excluding the test set, e.g., with Students test data, the model variant was trained on Flock, Zara, and ETH. We see that the test latents fall mostly within the training latents, providing further evidence for the method's generalisability.

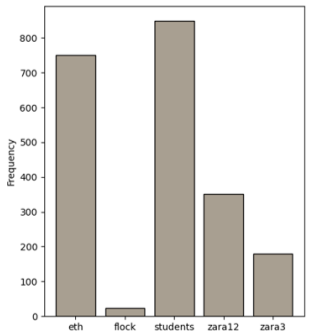


Fig. 1. Dataset sizes.

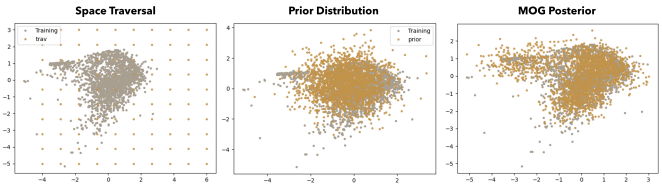


Fig. 2. Methods for sampling the latent space.

Latent Metric↓	k=2	k=3	k=4	k=5
MPACT	**1.873**	**1.867**	**1.488**	**1.262**
RVO	1.985	2.732	2.294	2.194
CCP	3.181	4.433	2.618	1.648

TABLE I
Number of latent clusters ablation.

B. Number of Latent Clusters

Additionally, we perform experiments for the number of latent clusters (k), with $k = 2, 3, 4, 5$ and report the latent metrics for MPACT, RVO, and CCP in Table I. We find that for all k, MPACT has the lowest value and is therefore ranked the closest to the real. The second best is found as the RVO, in all cases except for k=5, when CCP is ranked second. So, on average, we find a common trend among the different k, indicating that the findings are not highly sensitive to the

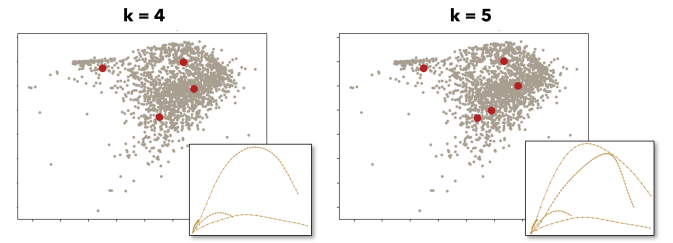


Fig. 3. Cluster centres in the latent space for k= 4,5 along with the respective inferred core paths.

number of clusters. Still, some changes still emerge. Having said that, the more cluster centres we have the more faithful the MoG will be to the data, but we want the cluster centres to also serve as core behaviours, and thus to contain diverse "enough" movements. So we trade-off k with core behaviours being distinct enough. Figure 3 shows the cluster centres for k=4,5 along with the inferred core behaviours (paths), indicating that the larger the k, the less distinct the core paths become. For a comparison independent of k, the graphical observation of the latent points of different simulators against the real data projections is preferable.

C. Inter-Cluster Feature Distributions

Figure 4 shows example trajectories from different clusters to help determine whether clusters capture meaningful behavioural differences. However, since the model aims at behavioural attributes and not trajectory shapes exclusively, we also test for inter-cluster differences through specific calculated features, namely: mean speeds, lengths, mean turning, straightness, and tortuosity, and present the findings in boxplots (Figure 10) showing that the different clusters, contain different distributions of features.

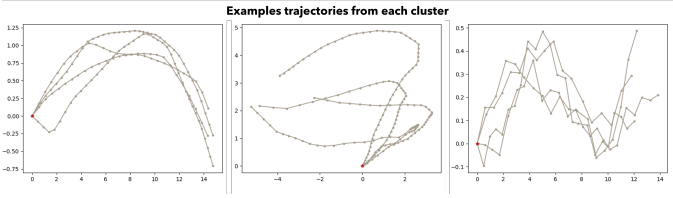


Fig. 4. Examples of intra-cluster trajectories.

IV. Complementary Results

Figure 8 plots the distribution of trajectory features for comparing the input training data, to the reconstructed data.

Figure 5 shows an additional two types of interesting data pairs apart from the slow-fast, namely the short-long, and high-low displacements' variability.

Figure 6 reveals another case of projections falling outside the learned MOG, representing latent embeddings of people instructed to move within an oval shape [4].

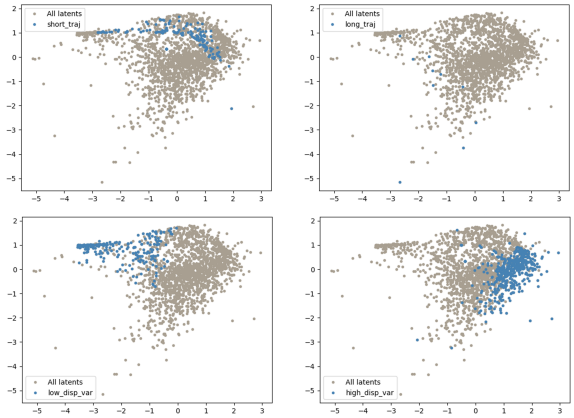


Fig. 5. Additional Latent Values of Interesting Datapoints. Overlay of all training latents and latents corresponding to: short/long paths (top), and low/high variability in the relative position distribution (bottom).

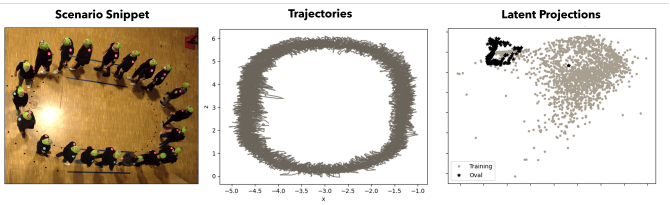


Fig. 6. Latent projections of people moving in a controlled oval scenario.

References

[1] J. Van den Berg, M. Lin, and D. Manocha, "Reciprocal velocity obstacles for real-time multi-agent navigation," in *2008 IEEE international conference on robotics and automation*, 2008.

[2] A. Panayiotou, T. Kyriakou, M. Lemonari, Y. Chrysanthou, and P. Charalambous, "Ccp: Configurable crowd profiles," in *ACM SIGGRAPH 2022 Conference Proceedings*, 2022.

[3] M. Lemonari, A. Panayiotou, T. Kyriakou, N. Pelechano, Y. Chrysanthou, A. Aristidou, and P. Charalambous, "Mpact: Mesoscopic profiling and abstraction of crowd trajectories," *Computer Graphics Forum*, 2025.

[4] A. K. Boomers, M. Boltes, J. Adrian, M. Beermann, M. Chraibi, S. Feldmann, F. Fiedrich, N. Frings, A. Graf, A. Kandler *et al.*, "Pedestrian crowd management experiments: A data guidance paper," *arXiv preprint arXiv:2303.02319*, 2023.

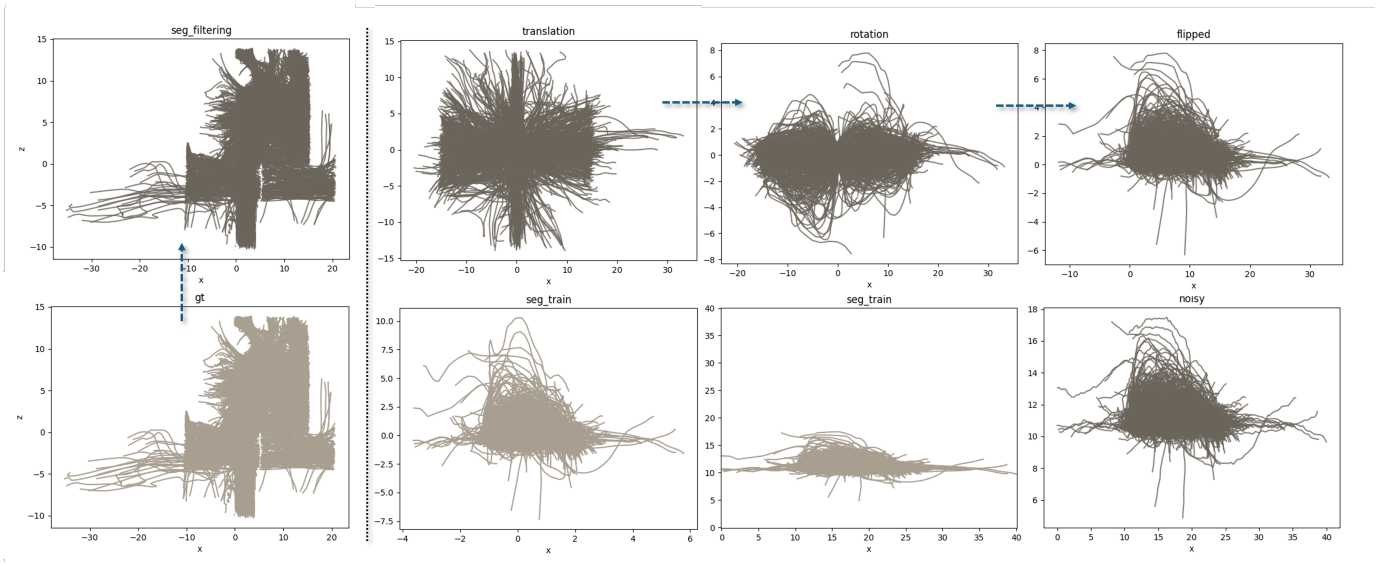


Fig. 7. Visualisation of the ablated preprocessing steps.

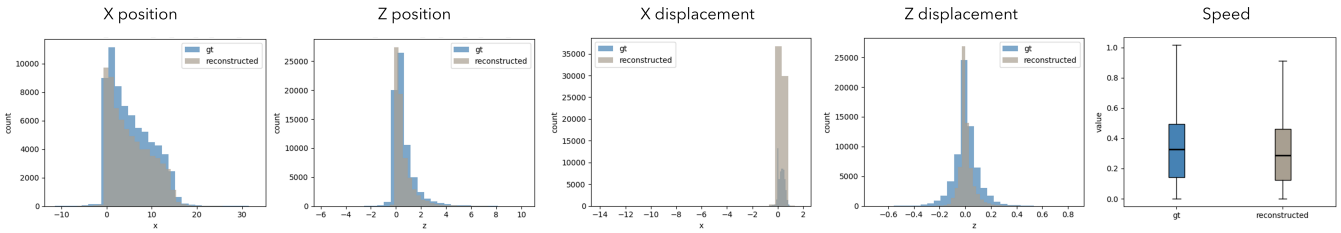


Fig. 8. Feature Plots of Training and Reconstructed Data. Histograms of positions and relative positions of x and z, and speed boxplot.

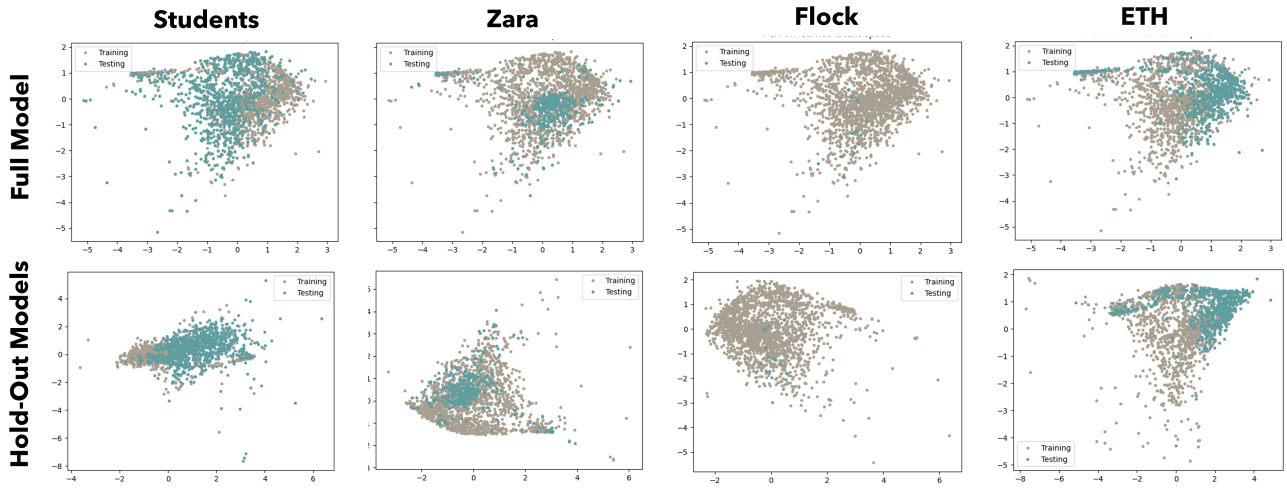


Fig. 9. Test data from Students, Zara, Flock, and ETH, projected onto the full model's latent space (top) and the latent spaces of model variants, trained on the datasets excluding the test set (bottom).

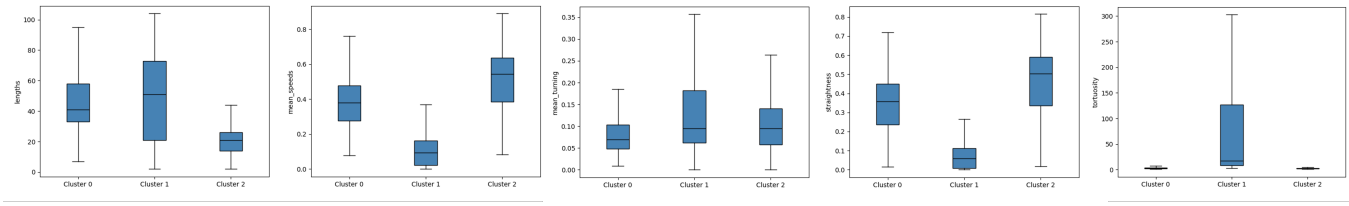


Fig. 10. Inter-cluster trajectory feature distributions for: mean speeds, path lengths, mean turning, straightness, and tortuosity.